HUMAN MOVEMENT 2 (MOTOR SKILLS DEVELOPMENT)

ACADEMIC

Grade 11/12

Curriculum Guide

HUMAN MOVEMENT 2
(MOTOR SKILLS DEVELOPMENT)
Prerequisite:
Human Movement 1 (Basic Anatomy in Sports and Exercise)

Course Description:

This course covers the fundamentals of motor skill development and its application in optimizing human movement. Learners will explore the classifications of motor skills and examine the stages of motor learning and motor control theories to understand their relevance in fitness and exercise. Through practical activities, learners will apply these concepts to enhance functional movement patterns and performance.

Elective: Academic
Time Allotment: 80 hours for one term

Content Standard:

- The learners demonstrate understanding of the fundamentals of motor skill development, motor learning stages, motor control theories, and movement assessment principles for optimizing functional movement performance in fitness and exercise.

Performance Standard:

- The learners evaluate basic functional movement patterns using motor learning principles, motor control theories, and contemporary movement assessment techniques to enhance performance.

<table>
<tr><th>LEARNING COMPETENCIES</th><th>CONTENT</th></tr>
<tr><td>The learners...
1. explain the different classifications of motor skills to optimize performance;</td><td>Classifications of Motor Skills
Gross Motor Skills vs. Fine Motor Skills
Discreet vs. Serial vs. Continuous Motor Skills
Open vs. Closed Skills
Overview of the Stages of Motor Skill Development</td></tr>
<tr><td>2. describe the factors influencing motor learning factors in relation to the stages of motor learning in fitness and exercise;</td><td><table>
<tr><th>Motor Learning Stages</th><th>Factors Affecting Motor Learning</th></tr>
<tr><td>Stage 1: The Cognitive Stage</td><td rowspan="3">Practice
Feedback
Attention
Teaching strategies
Cognitive stimulation
Social Factors
Individual Factors
Age</td></tr>
<tr><td>Stage 2: The Associative Stage</td></tr>
<tr><td>Stage 3: The Autonomous Stage</td></tr>
</table></td></tr>
<tr><td>3. apply motor control theories in relation to fitness and exercise;</td><td>The Motor Control Theories
Open and Close Loop Control Systems
Reflex Theory
Dynamical Systems Theory
Motor Programs
Hierarchical Theory</td></tr>
<tr><td>4. analyze movement patterns using measurement techniques to enhance performance; and</td><td>Measuring and Analyzing Human Movement<table>
<tr><th>Suggested Activities</th><th>Methods of Analysis</th></tr>
<tr><td>Movement Screening</td><td>Qualitative Techniques Using Video Analysis</td></tr>
</table></td></tr>
</table>

LEARNING COMPETENCIES	CONTENT	
	Movement Screening *(Suggested movement screens: Deep Squat, Hurdle Step, In-line Lunge, Active Straight-Leg Raise, Trunk Stability Push-Up, Rotary Stability, and Shoulder Mobility)* Basic Movement Patterns • Push and Pull • *Vertical* • *Horizontal* • *Diagonal* • *Rotational*	• Qualitative Techniques Using Video Analysis • Quantitative Techniques Using Free Software Applications (e.g., Kinovea, PhysioU Gait App, OpenCap, Dartfish, Functional Movement Test (FMT) App, Symmio Self-Screen App) • Software Applications for Movement Assessment
5. evaluate current trends in movement assessment to optimize performance.	Current Trends in Movement Assessment	

GLOSSARY

Adaptive movement — The ability to modify motor responses based on environmental changes or sensory feedback to achieve a desired goal.

Associative Stage — The stage of motor learning in which movements become more fluid, reliable, and efficient. The learner refines movement patterns and combines smaller movement skills into more complex actions.

Automatic Movement Theory — A theory that explains the neural and functional basis for movements performed with minimal conscious effort. These movements are often habitual, repetitive, or acquired through extensive practice.

Autonomous Stage	The final stage of motor learning is in which movements are accurate, consistent, and efficient. The learner performs the movement with little or no conscious cognitive processing.
Cognitive Stage Theory	The initial stages of motor learning are slow, inconsistent, and inefficient. The learner processes information to understand the requirements of the movement.
Current Trends	Developments and innovations that are occurring or being used at present. In movement studies, these include advancements in technology, methodologies, and applications in physical performance, rehabilitation, and sports science.
Fine Motor Skills	It refers to physical skills that involve small muscles and hand-eye coordination. Movements are more controlled and precise, and often a well-developed pincer grip is needed.
Functional Movement Patterns	It refers to fundamental movements that form the basis of daily activities and physical tasks, such as squatting, lunging, pushing, pulling, and bending. These patterns are essential for maintaining balance, stability, and mobility while performing various tasks. Assessing these patterns helps identify potential weaknesses, imbalances, or inefficient biomechanics that could lead to injuries or reduced performance.
Functional Movement Screen (FMS)	It evaluates an individual's ability to perform seven fundamental movements, such as deep squat and inline lunge. These tests highlight areas requiring improvement in mobility, stability, or motor control, enabling targeted interventions through corrective exercises. This approach is critical for athletes, rehabilitation, and anyone seeking to enhance functional fitness.
Human Movement	It is the movement produced by the human body due to the contraction of muscles and bending of bone joints. Human movements are controlled by the nervous system. Hence, the human movement incorporates the use of muscles, ligaments, joints, and bones. Kinesiology is the field that deals with the study of human movements.
Human Movement Analysis	It is the study of human motion using various techniques such as biomechanics, kinesiology, and motion capture technology. A method of analysis refers to the systematic approach used to study, interpret, and evaluate a subject, phenomenon, or process.
Human Movement System	It is a system of physiological organ systems that interact with them. to produce movement of the body and its parts.

Human Physiology	It studies the “nature” of the human body, nature in the sense of how structures at different levels work. Physiology focuses on function, or how structures at different levels work.
Motor Control	It is a complex process involving the coordinated contraction of muscles due to the transmission of impulses sent from the motor cortex to its motor units. It is the process of initiating, directing, and grading purposeful voluntary movement.
Motor learning	It is the brain’s way of committing automatic reaction to memory by practicing a skill or action repeatedly until it’s ingrained in the brain and central nervous system. The repetition of motor learning enables people to permanently change their brains so they can automatically react in each situation without having to think on a conscious level.
Movement Assessment	It evaluates a person's ability to perform functional movements, often focusing on mobility, stability, strength, coordination, and overall movement patterns. It is used in various fields, such as physical therapy, sports science, and fitness training, to identify movement deficiencies, imbalances, or risks of injury.
Nervous System	The body system is responsible for the initiation and regulation of vital body functions, sensation, and movement.
Qualitative Analysis of Movement	The systematic observation and evaluation of movement quality to identify inefficiencies and improve performance. Examples include coaching through video replay using applications such as Kinovea.
Quantitative Analysis of Movement	The measurement and evaluation of movement using numerical data such as speed, joint angles, distance, force, and time.
Reflex Theory	A theory that explains how reflex responses to stimuli influence movement. It posits that complex movements and behaviors result from combinations of simpler reflexes. Sensory input plays a vital role in this theory because it triggers reflexive responses.
Software Application	A program or set of programs designed to perform specific tasks. In human movement analysis, software applications are used to assess, record, and analyze movement patterns.

Stability	The ability of the body to maintain postural equilibrium and support joint integrity during movement.
Torso	The main part of the body contains the chest, abdomen, pelvis, and back. It is also called the trunk.
Video Analysis	A technique used to observe, record, and evaluate movement patterns and athletic performance to provide feedback and improve technique.
Voluntary movement	It refers to intentional and goal-directed actions initiated and controlled by the higher centers of the brain, particularly the motor cortex, basal ganglia, and cerebellum. This theory explains how the brain plans, initiates, and executes purposeful actions, such as picking up an object or writing.

REFERENCES

BPT, U. T. (2024, June 24). *Human anatomy terminology*. Muscle and Motion. https://www.muscleandmotion.com/human-anatomy-terminology/

Contemporary issues and theories of motor control, motor learning, and neuroplasticity. (2015, April 9). *Clinical Gate*. https://clinicalgate.com/contemporary-issues-and-theories-of-motor-control-motor-learning-and-neuroplasticity/

Emmert, W. (2010). *Functional movement screen*. American College of Sports Medicine. https://www.acsm.org/docs/default-source/regional-chapter-individual-folders/northland/nacsm--wes-e--fms9a9b0c1f5032400f990d8b57689b0158.pdf

Energy system. (n.d.). *Open Risk Manual*. https://www.openriskmanual.org/wiki/Energy_System

Evans, W. J., & Lambert, C. P. (2007). Physiological basis of fatigue. *American Journal of Physical Medicine & Rehabilitation, 86*(1), S29–S46.
https://doi.org/10.1097/phm.0b013e31802ba53c

Gross and fine motor skills. (n.d.). *Early Movers*.
https://www.earlymovers.org.uk/motor-skills

Growth, maturation and development. (n.d.). *World Rugby Passport*.
https://passport.world.rugby/conditioning-for-rugby/introduction-to-conditioning-children/long-term-athlete-development/growth-maturation-and-development/

Human movement. (2021, October 22). *Bartleby*.
https://www.bartleby.com/subject/science/biology/concepts/human-movement

Latash, M. L., Levin, M. F., Scholz, J. P., & Schöner, G. (2010). Motor control theories and their applications. *Motor Control, 14*(2), 1–43.
https://pmc.ncbi.nlm.nih.gov/articles/PMC3017756/

Mobility and stability: Joint functions when we move. (n.d.). *National Academy of Sports Medicine*.
https://blog.nasm.org/certified-personal-trainer/mobility-and-stability-joint-functions-when-we-move

Motor control and motor learning. (2018, April 26). *Trek Education*.
https://exercise.trekeducation.org/motor-learning/

Sahrmann, S. (n.d.). *The human movement system: Our professional identity*. Program in Physical Therapy, University of Toledo.
https://www.utc.edu/sites/default/files/2020-12/annualforum_2017_by_sahrmann_012617.pdf

The hypothesis that reflexes form the basic language of the motor program permits simple, flexible specifications of voluntary movements. (n.d.). *JSTOR*.
https://www.jstor.org/stable/27843368